

Name: Mewada Maitrri Maishkumar

Topic: Linux File System Management, Permissions, Ownership, Links, and User/Group Administration

1. Open a terminal and use the 'ls', 'cd', and 'pwd' commands to navigate to the /home directory, then list all user folders present. Take a screenshot of your terminal showing the commands and output.

A screenshot of a Linux terminal window. The window title is "Red Hat Enterprise Linux ...". The terminal shows the following commands and output:

```
[maitrri@localhost ~]$ pwd
/home/maitrri
[maitrri@localhost ~]$ cd /home
[maitrri@localhost home]$ pwd
/home
[maitrri@localhost home]$ ls
maitrri
[maitrri@localhost home]$
```

The terminal window has a dark background with a light blue border. The top bar shows "Activities" and "Terminal" tabs, and the date "Aug 7 13:37". The terminal window itself has a title bar with "maitrri@localhost:/home" and search, menu, and close buttons.

2. Create a new directory called 'myplaylist' inside your home folder, then use 'touch' to create three empty files named spotify.txt, youtube.txt, and wynk.txt inside it. Set the permissions so only you (the owner) can read and write these files, but others cannot

access them.

Hint: Use 'chmod 600' on each file after creating them.

⇒ Go to your home directory:

```
cd ~
```

Create directory:

```
mkdir myplaylist
```

Enter directory:

```
cd myplaylist
```

Create three empty files:

```
touch spotify.txt youtube.txt wynk.txt
```

Set permissions:

```
chmod 600 spotify.txt
```

```
chmod 600 youtube.txt
```

```
chmod 600 wynk.txt
```

Check permissions:

```
ls -l
```

3. Change the ownership of the 'myplaylist' directory and all its files to another user on your system using the 'chown' command. If you are the only user, create a new user first.

Hint: Use 'sudo adduser newusername' to create a user if needed.

⇒ First create a new user (if required):

```
sudo adduser musicuser
```

Change ownership of directory and files:

Go back to home

```
cd ~
```

Run:

```
sudo chown -R musicuser myplaylist
```

Check ownership:

```
ls -l
```

4. Inside your home directory, create a symbolic link named 'favmusic' that points to the spotify.txt file you created earlier. Then, create a hard link named 'spotify_hardlink' to the same file. Show the output of 'ls -l' and 'ls -li' to demonstrate the difference between the links.

⇒ Go to home directory:

```
cd ~
```

Create symbolic link:

```
ln -s myplaylist/spotify.txt favmusic
```

Create hard link:

```
ln myplaylist/spotify.txt spotify_hardlink
```

Show symbolic link:

```
ls -l
```

Show inode numbers:

```
ls -li
```

5. Suppose you want to share your 'myplaylist' folder with a group of friends on the same Linux system. First, create a new group called 'musiclovers' using the 'groupadd' command, add your user to this group, and then change the group ownership of 'myplaylist' to 'musiclovers' using 'chgrp'. Set the permissions so that only group members can read and write files in this directory, but others cannot.

Hint: Use 'chmod 770' on the directory after changing the group.

⇒ Create a new group:

```
sudo groupadd musiclovers
```

Add your user to group:

```
sudo usermod -aG musiclovers username
```

Change group ownership:

```
sudo chgrp -R musiclovers myplaylist
```

Set directory permissions:

```
chmod 770 myplaylist
```

Check:

```
ls -ld myplaylist
```